

Brian Yoon

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Education

University of Illinois at Urbana-Champaign	Aug 2024 - May 2025
<i>Master of Computer Science GPA 3.97/4.0</i>	
University of Illinois at Urbana-Champaign	Aug 2023 - Aug 2024
<i>Graduate Certificate Illinois Computing Accelerator for Non-Specialists (iCAN) GPA: 3.93/4.0</i>	
University of Illinois at Urbana-Champaign	Aug 2019 - Dec 2022
<i>B.S in Molecular and Cellular Biology GPA 3.98/4.0</i>	

Skills

- **Languages:** Python, Java, C#, JavaScript/TypeScript, SQL
- **Backend:** Spring Boot, FastAPI, Node.js, Express.js, REST APIs
- **Frontend:** React, HTML, CSS
- **AI/ML:** AI APIs, AI-Assisted Development, Applied Machine Learning, User-Centered Machine Learning
- **Databases:** PostgreSQL, MySQL, MongoDB, Neo4j
- **Tools/Cloud:** Git, Docker, Linux, Postman, Bruno, JUnit
- **Cloud:** Google Cloud Platform
- **Testing:** Unit Testing, API Testing
- **Data & Graph:** Pandas, NumPy, NetworkX, Graph/Network Modeling, Data Pipelines (ETL)

Projects

Spring Boot Task Manager (Spring Boot Java Spring Data JPA):	May 2026- June 2026
• Designed and engineered a production-ready, multi-entity task management REST API using a Three-Tier Layered Architecture (Controller, Service, Repository) built with Java and Spring Boot. • Leveraged Spring Data JPA and Hibernate to model complex relational data structures, incorporating foreign key constraints and lazy-loading strategies (FetchType.LAZY) to optimize database performance. • Improved maintainability by implementing DTO-based data transfer, custom object mapping, and centralized exception handling for standardized API responses. • Integrated Spring Boot Actuator to expose critical telemetry, request-routing maps, and live application health metrics.	
AI Room Decoration Helper (OpenAI API JavaScript/TypeScript React):	April 2025 - May 2025
• Built an AI-powered room decoration application integrating AI APIs to generate room design recommendations and assist users with personalized decoration ideas. • Designed the application workflow to incorporate AI-generated outputs into an interactive user experience.	
PrairieLearn Venn Diagram (JavaScript Python HTML CSS):	Oct 2024 - Dec 2024
• Designed and developed a new interactive question type for University of Illinois at Urbana-Champaign's PrairieLearn platform using JavaScript, Python, HTML, and CSS in a team environment. • Built an interactive Venn diagram interface allowing users to dynamically create, label, move, and shade circles for automated assessment in Civil Engineering coursework.	
PiggyBank (JavaScript MySQL HTML CSS):	July 2024-Aug 2024
• Led the development of a college student budgeting platform as Team Leader and Full-Stack Developer in a collaborative, Agile-style team environment, iteratively delivering features based on user needs. • Utilized Google Cloud Platform for scalable hosting and built the backend with Node.js and Express.js. Designed a MySQL database for secure financial data management and delivered an intuitive user experience for effective budget tracking.	

Professional/Work Experience

Computational Research Intern:	May 2024 - Aug 2024
• Engineered graph and network models in Python and PostgreSQL to analyze complex biological/medical citation datasets, modeling relationships across 20,000+ interconnected nodes and edges within a Linux virtual environment accessed via SSH. • Constructed automated data extraction and transformation pipelines using NetworkX and SQL, performing graph query analysis and entity linkage to identify high-impact domain publications. • Utilized the CM Pipeline library for in-depth analysis of network data. Collaborated with Dr. Diane Maresco-Pennisi to identify and evaluate the top ten most influential research publications, employing quantitative methods to measure their relevance and impact.	
Office of Minority Students Affairs Academic Tutor:	Jan 2021- Dec 2022
• Enhanced leadership, management, and communication skills by tutoring university students from minority groups in diverse academic subjects, including biology, chemistry, and physics.	

Activities

Computer Science Graduate Student Organization (CSGSO):	Aug 2024 - Aug 2025
• Collaborated with peers and faculty to enhance the academic and social experience for CS graduate students at UIUC, organizing visit weekends, facilitating communication with leadership, and advocating for student needs.	